

Project Initialization and Planning Phase

Date	30 june 2026
Team ID	xxxxxxx
Project Title	Credit card approval prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective is to revolutionize the credit assessment and card provisioning lifecycle by implementing advanced machine learning classification techniques, ensuring instantaneous, unbiased, and exceptionally accurate risk assessment loops.
Scope	The project comprehensively automates the verification boundary by analyzing applicant demographic features from application_record.csv alongside granular chronological repayment matrices within credit_record.csv. It replaces slow manual checks with a responsive, live local web inference dashboard wrapper..
Problem Statement	
Description	Traditional credit card evaluations rely heavily on manual verification checklists and rigid linear rule systems that fail to intercept non-linear socioeconomic feature interactions. Furthermore, joining disparate tables line-by-line introduces significant operational backlogs, severe administrative delays, cognitive fatigue, and human decision bias.

Impact	Addressing these inefficiencies will drop verification cycle durations dramatically, mitigate loan default patterns (Non-Performing Assets), optimize operational resource expenditure, and establish a high-satisfaction financial inclusion model for applicants..
Proposed Solution	
Approach	Employing specialized supervised machine learning classification algorithms (such as Random Forest Ensembles with class-balancing matrices) to automatically aggregate historical database silhouettes, handle structured missing data imbalances, and map variables instantly to definitive risk verdicts.
Key Features	- Automated data consolidation pipeline pairing socioeconomic profiles with rolling monthly billing tracks on explicit unique ID indexes. • Structural missing data imputation handling the 30.5% unrecorded space in categorical values (OCCUPATION_TYPE) without loss of applicant profiles. • Embedded minority-class weighting matrix adjustments to correct severe target profile imbalance. • Real-time, zero-latency local prediction inference through an accessible, web-based layout dashboard using simple Bootstrap 5 controls.



	- Real-time decision-making for quicker loan approvals. - Continuous learning to adapt to evolving financial landscapes.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU processing core counts and execution platforms for baseline model training.	Intel Core i5/i7 or AMD Ryzen 5/7 equivalent (4+ Physical Cores)

Memory	Volatile RAM allocation required to manipulate merged Pandas DataFrames in active memory.	8 GB RAM (Minimum) / 16 GB RAM (Recommended)
Storage	High-speed permanent storage space for local data arrays, source tables, and serialized pipeline binaries.	256 GB SSD (Standard local storage is sufficient for dataset profiles)
Software		
Frameworks	core server architecture driving the interface routing logic.	Python WSGI Server Framework: Flask (v3.0+)
Libraries	core server architecture driving the interface routing logic.	scikit-learn, pandas, numpy, matplotlib, seaborn, pickle
Development Environment	Interactive development engine platforms for pipeline script compilation and local sandbox testing.	Jupyter Notebook (Exploratory Phase) & PyCharm / VS Code (Web Deployment Build)
Data		
Data	Data source extraction channels, observation row volumetric profiles, and local file storage formatting.	Comprehensive Credit Card Approval Dataset (consisting of linked tracking arrays in .csv format: application_record and credit_record).